

5.1 For $V_{DD}=1.8$ V and using $I_{REF}=100$ μ A, it is required to design the circuit of Fig 1 to obtain an output current whose nominal value is 100 μ A. If Q_1 and Q_2 are matched with channel lengths of 0.5 μ m, channel widths of 5 μ m, $V_{TH}=0.4$ V , and $\mu_n C_{OX}= 500$ μ A/V².

(a) Find R

(b) What is the lowest possible value of V_O ?

(c) Assuming that for this process technology the $\lambda=0.4V^{-1}$, find the output resistance of the current source.

(d) Also, find the change in output current resulting from a +0.5V change in V_O .

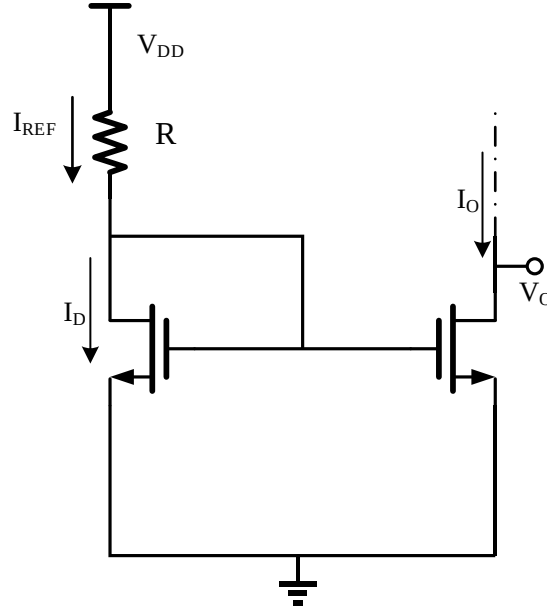


Fig 1

(a)

$$I_D = \frac{1}{2} \mu_n C_{oc} \left(\frac{W}{L} \right) V_{od}^2 \implies V_{od} = 0.2 \text{ V}$$

$$V_{GS} = V_{TH} + V_{od} = 0.6 \text{ V} \quad (1-1)$$

$$R = \frac{V_{DD} - V_{GS}}{I_{REF}} = 12 \text{ k}\Omega$$

(b)

$$V_{o,\min} = V_{od} = 0.2 \text{ V} \quad (1-2)$$

(c)

$$r_o = \frac{1}{\lambda I_D} = \frac{1}{0.4 \times 100 \times 10^{-6}} = 25000 \quad (1-3)$$

(d)

$$\Delta I = \frac{\Delta V_O}{r_O} = \frac{0.5}{25k} = 20 \text{ } \mu\text{A} \quad (1-4)$$

5.2 A PMOS current mirror consists of three PMOS transistors, one diode connected and two used as current outputs. All transistors have $|V_{TH}|=0.6V$, $\mu_p C_{ox}=100\mu A/V^2$, and $L=1\mu m$ but three different widths, namely, $10\mu m$, $20\mu m$, and $40\mu m$. When the diode-connected transistor is supplied from a $100\mu A$ source, how many different output currents are available? Repeat with two of the transistors diode connected and the third use to provide current output. For each possible input-diode combination, give the values of output currents and of the V_{SG} that results. ($\gamma=0$, $\lambda=0 V^{-1}$)

$V_{SG}(V)$	Q1(10)(μA)	Q2(20) (μA)	Q3(40) (μA)
Diode Q1(1.05)	100	200	400
Diode Q2(0.92)	50	100	200
Diode Q3(0.82)	25	50	100
Diode Q1 Q2(0.86)			133
Diode Q1 Q3(0.80)		40	
Diode Q2 Q3(0.78)	16.7		

5.3 The circuit of Fig 2 is designed with $\mu_n C_{ox}(W/L)_n = 4 \text{ mA/V}^2$ and $I_{in}=36 \text{ mA}$, determine the minimum allowable voltage V_{OUT} , ($\lambda=0.1 \text{ V}^{-1}$)

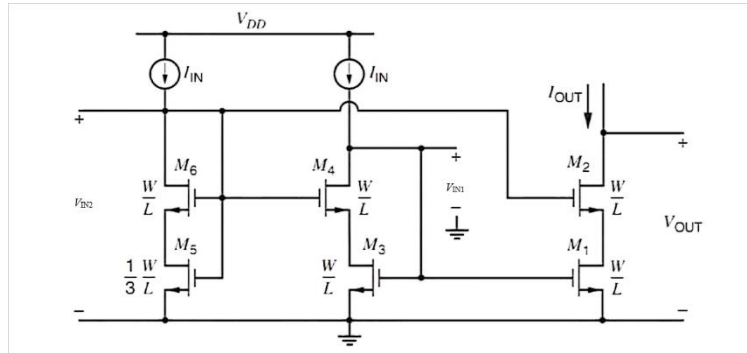


Fig 2

$$I_{in} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right) V_{od}^2 \Rightarrow V_{od} = 0.3 \text{ V} \quad (3-1)$$

$$V_{out} \geq V_{in2} - V_{TH} = V_{GS5} - V_{TH} \geq (V_{GS4} + V_{od}) - V_{TH} = 2V_{od} = 0.6 \text{ V}$$

5.4 In the common-gate amplifier circuit of Fig 3, Q_2 and Q_3 are matched. $\mu_n C_{ox}(W/L)_n = \mu_p C_{ox}(W/L)_p = 4 \text{ mA/V}^2$, and all transistors have $|V_{TH}| = 0.8 \text{ V}$ and $\lambda = 1/20 \text{ V}^{-1}$. The signal v_{sig} is a small sinusoidal signal with no dc component. ($\gamma=0$)

- Neglecting the effect of channel length modulation, find the dc drain current of Q_1 and the required value of V_{BIAS} .
- Find the values of g_{m1} and r_o for all transistors.
- Find the value of R_{in} .
- Find the value of R_{out} .
- Calculate the voltage gains v_o/v_i and v_o/v_{sig} .
- How large can v_{sig} be (peak-to-peak) while maintaining saturation-mode operation for Q_1 and Q_2 ?

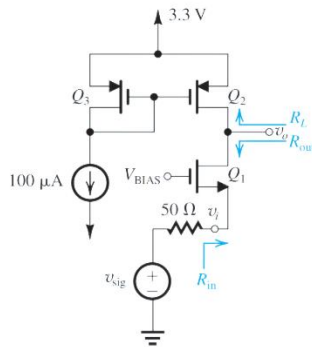


Fig 3

(a)

$$I_{D1} = I_{D2} = I_{D3} = 100 \mu A, R_S = 50 \Omega$$

$$I_{D1} = \frac{1}{2} \mu_n C_{oc} \left(\frac{W}{L} \right) V_{od1}^2 \implies V_{od1} = 0.224 V \quad (4-1)$$

$$V_{BIAS} = V_{GS1} + I_{D1} R_S = V_{od1} + V_{TH} + I_{D1} R_S = 1.029 V$$

(b)

$$g_{m1} = \frac{2I_{D1}}{V_{od1}} = \frac{2 \times 0.1}{0.224} = 0.892857 mS$$

$$r_o = \frac{1}{\lambda I_D} = \frac{20}{0.1 \times 10^{-3}} = 200000 \Omega \quad (4-2)$$

(c)

$$R_L = r_{o2}$$

$$R_{in} = \frac{R_L + r_{o1}}{1 + g_{m1} r_{o1}} = \frac{200000 \times 2}{1 + 0.892857 \times 200} = 2227.53 \Omega \quad (4-3)$$

(d)

$$R_{out} = R_s + r_{o1} + g_m r_{o1} R_s = 200.05 + 0.892857 \times 200 \times 0.05 = 208.979 k\Omega \quad (4-4)$$

(e)

$$A_V = \frac{v_o}{v_i} = \frac{1 + g_m r_o}{r_o + R_L} R_L = \frac{1 + 0.892857 \times 200}{400000} \times 200000 = 89.7857$$

$$\frac{v_i}{v_{sig}} = \frac{R_{in}}{R_{in} + R_S} = \frac{2227.53}{2227.53 + 50} = 0.978046 \quad (4-5)$$

$$\frac{v_o}{v_{sig}} = \frac{v_o}{v_i} \frac{v_i}{v_{sig}} = 89.7857 \times 0.978046 = 87.8145$$

(f)

$$v_{o,max} = V_{DD} - V_{od1} = 3.3 - 0.224 = 3.076 V$$

$$v_{o,min} = V_{BIAS} - V_{TH} = 0.229 V$$

$$v_{sig} = \frac{v_{o,max} - v_{o,min}}{A_V} = \frac{3.076 - 0.229}{89.7857} = 0.0317088 = 32 mV$$

5.5 Consider the Wilson MOS mirror of Fig 4(a) for the case of all transistors identical, with $W/L=10$, $\mu_n C_{ox} = 0.4 \text{ mA/V}^2$, $V_{TH} = 0.5 \text{ V}$, and $\lambda = 1/18 \text{ V}^{-1}$. The mirror is fed with $I_{REF} = 180 \text{ }\mu\text{A}$.

- Obtain an estimate of V_{od} and V_{GS} at which the three transistors are operating, by neglecting the Early effect.
- Noting that Q_1 and Q_2 are operating at different V_{DS} , obtain an approximate value for the difference in their currents and hence determine I_O .
- To eliminate the systematic error between I_O and I_{REF} caused by the difference in V_{DS} between Q_1 and Q_2 , a diode-connected transistor Q_4 can be added to the circuit as shown in Fig 4(b). What do you estimate I_O now to be?
- What is the minimum allowable voltage at the output node of the mirror?
- Convince yourself that Q_4 will have no effect on the output resistance of the mirror. Find R_o .
- What is the change in I_O (both absolute value and percentage) that results from $\Delta V_O = 1 \text{ V}$?

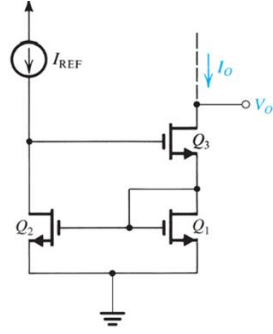


Fig 5(a)

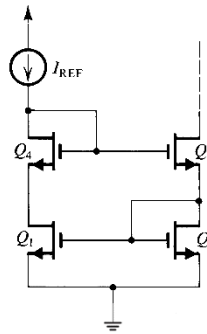


Fig 4(b)

(a)

$$V_{od} = \sqrt{\frac{2I_D}{\mu_n C_{ox} \frac{W}{L}}} = \sqrt{\frac{2 \times 180 \times 10^{-6}}{0.4 \times 10^{-3} \times 10}} = 0.3$$

$$V_{GS} = V_{od} + V_{th} = 0.8 \text{ V}$$

(b)

$$V_{DS2} = 2V_{GS} = 1.6 \text{ V}$$

$$\frac{I_O}{I_{REF}} = \frac{1 + \lambda V_{DS1}}{1 + \lambda V_{DS2}} = \frac{1 + 1/18 \times 0.8}{1 + 1/18 \times 1.6} = 0.959184$$

$$I_O = 0.959 \times 180 = 172.62 \text{ }\mu\text{A}$$

(c)

$$I_O = I_{REF}$$

(d)

$$V_{D3} > V_{GS2} + V_{GS3} - V_{TH} = 2V_{od} + V_{th} = 1.1 \text{ V}$$

(e)

$$R_O = g_{m2} r_{o3} r_{o2} = \frac{2 \times 180 \times 10^{-6}}{0.3} \times \frac{1}{1/18 \times 180 \times 10^{-6}} \times \frac{1}{1/18 \times 180 \times 10^{-6}} = 1.2 \times 10^7 \Omega$$

(f)

$$\Delta I = \frac{\Delta V_O}{R_O} = \frac{1}{1.2 \times 10^7} = 8.33333 \times 10^{-8} \text{ A}$$

5.6 The circuit of Fig 6 is designed with $(W/L)_{1,2} = 20/0.5$, $(W/L)_{3,0} = 60/0.5$, and $I_{REF} = 100 \mu A$.

(a) Determine V_X and the acceptable range of V_b . ($\lambda=\gamma=0$)

(b) Estimate the deviation of I_{out} from $300 \mu A$ if the drain voltage of M_3 is higher than V_X by 1 V.

($\gamma=0$, $\lambda=0.1 V^{-1}$, $\mu_n C_{ox} = 1.34225 \times 10^{-4}$)

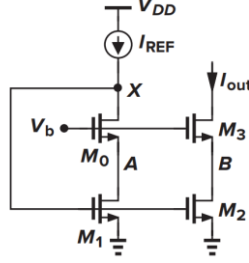


Fig 6

(a)

$$V_X = V_{TH1} + \sqrt{\frac{2I_{REF}}{\mu_n C_{ox}(W/L)_1}} \approx 0.893V$$

为保证 M0 饱和, $V_b < V_X + V_{TH0} = 1.593V$

为保证 M1 饱和, $V_b > V_{GS0} + V_{OD1} = V_{TH0} + \sqrt{\frac{2I_{REF}}{\mu_n C_{ox}(W/L)_0}} + \sqrt{\frac{2I_{REF}}{\mu_n C_{ox}(W/L)_1}} \approx 1.086V$

(b)

$$r_{o2} = r_{o3} = \frac{1}{\lambda \cdot 3I_{REF}} = \frac{1}{0.1 \times 3 \times 100 \times 10^{-6}} = 33333.3 \Omega$$

$$g_{m3} = \sqrt{2\mu_n C_{ox} \left(\frac{W}{L}\right) (3I_{REF})} = \sqrt{6 \times 1.34225 \times 10^{-4} \times 100 \times 10^{-6} \times 120} = 0.00310873 S \quad (5-1)$$

$$\Delta I_{out} = \frac{\Delta V}{g_{m3} r_{o2} r_{o3}} = \frac{1}{0.00310873 \times 33333.3^2} = 2.89508 \times 10^{-7}$$

5.7 The circuit of Fig 7 is designed with $(W/L)_{1-4} = 50/0.5$ and $I_{SS} = 2I_1 = 0.5$ mA.

(a) Calculate the small-signal voltage gain.

(b) Determine the maximum output voltage swing if the input CM level is 1.3 V.

($\lambda_n = 0.1$, $\lambda_p = 0.2$, $|V_{Tn}| = 0.7$ V, $\gamma = 0$, $\mu_n C_{ox} = 1.34225 \times 10^{-4}$ $\mu_p C_{ox} = 0.3835 \times 10^{-4}$)

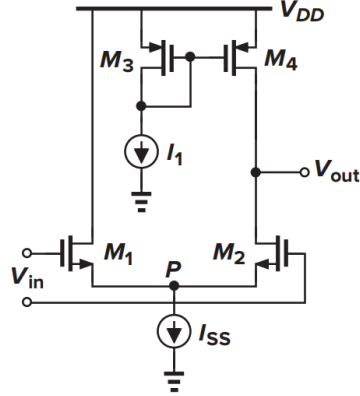


Fig 7

(a)

电路不对称，不能用半边等效电路的方法，具体求解课本见 P.133

$$A_v \approx -\frac{g_{m1}}{2}(2r_{o2} || r_{o4}) \approx -20.7$$

(b)

为使得 M1, M2 饱和, $V_{out} > V_{in,CM} - V_{THN} = 0.6V$

为使得 M4 饱和, $V_{out} < V_{DD} - V_{OD4} = V_{DD} - \sqrt{\frac{2I_1}{\mu_p C_{ox} \left(\frac{W}{L}\right)_4}} \approx 2.639V$

$$V_{out,sw} = 2.639V - 0.6V = 2.039V$$

5.8 Consider the circuit of Fig 8 with $(W/L)_{1-5} = 50/0.5$ and $I_{D5} = 0.5$ mA.

(a) Calculate the deviation of V_{out} from V_F if $|V_{TH3}|$ is 1 mV less than $|V_{TH4}|$.

(b) Determine the CMRR of the amplifier.

($\lambda_n = 0.1$, $\lambda_p = 0.2$, $|V_{TH4}| = 0.8$ V, $\gamma = 0$, $\mu_n C_{ox} = 1.34225 \times 10^{-4} \mu_p C_{ox} = 0.3835 \times 10^{-4}$)

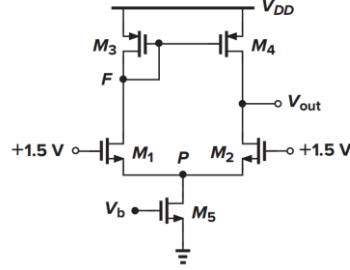


Fig 8

(a)

$$|V_{TH3}| = |V_{TH4}| - 1\text{mV}, |V_{TH4}| = 0.8\text{V}$$

$$I_{D3} = \frac{I_{D5}}{2} = \frac{1}{2} \mu_p C_{ox} (W/L)_3 (V_{DD} - V_F - |V_{TH3}|)^2 (1 + \lambda(V_{DD} - V_F))$$

$$I_{D4} = \frac{I_{D5}}{2} = \frac{1}{2} \mu_p C_{ox} (W/L)_4 (V_{DD} - V_F - |V_{TH4}|)^2 (1 + \lambda(V_{DD} - V_{out}))$$

$$V_F \approx 1.8748\text{V} \quad V_{out} \approx 1.837\text{V}$$

$$V_F - V_{out} = 37.7\text{mV}$$

(b)

$$A_{dm} = g_{m1,2} (r_{o1,2} || r_{o3,4})$$

$$A_{cm} = -\frac{1}{1 + 2g_{m1,2}r_{o5}} \times \frac{g_{m1,2}}{g_{m3,4}}$$

$$CMRR = \left| \frac{A_{dm}}{A_{cm}} \right| = (1 + 2g_{m1,2}r_{o5}) g_{m3,4} (r_{o1,2} || r_{o3,4}) \approx 1931.7$$